

ENTERPRISE DESIGN DOCUMENT

FacilityDesk AI-Powered QR Facilities Ticketing & Management Platform

Organisation: Bayer | **Function:** RE, Commercial Area & Facilities Management |
Version: 1.0

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Abstract. This document describes the end-to-end design of **FacilityDesk**, an enterprise-grade, AI-powered facilities ticketing platform for Bayer's office network across South Asia. Employees scan location-specific QR codes posted on office floors to raise maintenance complaints in plain English. A generative AI layer structures, categorises, prioritises, and auto-routes each ticket to the correct maintenance team. Facility managers track ticket lifecycle through a multi-stage workflow with SLA enforcement, automated escalation, WhatsApp and email notifications to technicians, and a Microsoft Teams-based feedback collection flow. An admin dashboard provides QR generation, cross-office analytics, and AI-generated feedback reports by day, month, and year. Azure Active Directory authenticates all users; no mock data is used in production.

PROJECT ORIGIN & MANDATE

This use case has been defined and assigned by the
Real Estate, Commercial Area & Facilities Management Team, Bayer

This platform was commissioned within Bayer's **RE, Commercial Area & Facilities Management** function. The requirements, ticket workflows, escalation rules, SLA definitions, and notification configurations described herein reflect the operational mandate of the facilities management team to modernise and automate the floor-level complaint resolution process across all Bayer office locations in South Asia.

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Organisation	Function	Point of Contact
Bayer	RE, Commercial Area & Facilities Management	Gandharv Chauhan
		Head – RE, Com Ar Admin & Fac Mgmt

1 Use Case Definition

1.1 Business Context

Bayer operates multiple office locations across South Asia — including Thane, Mumbai, and other regional offices — housing hundreds of employees across multiple floors. Each floor has ongoing operational needs: furniture repairs, plumbing issues, electrical faults, HVAC failures, housekeeping requests, IT infrastructure problems, and safety concerns. Currently, employees report these issues informally through phone calls, corridor conversations, or ad-hoc emails to facility staff — resulting in lost requests, slow resolution, no audit trail, and no visibility into recurring problem areas.

1.2 Problem Statement

There is no structured, trackable mechanism for employees to raise facility complaints. Requests are lost, duplicated, or misrouted. Facility managers have no real-time view of open issues. Resolution time is unmeasured. There is no feedback loop to assess service quality. Senior management has no data to make decisions about maintenance investment across offices.

1.3 Target State & Elevated Solution

FacilityDesk transforms this into an intelligent, end-to-end digital platform:

- a) **QR-first submission:** unique, location-aware QR codes posted at every zone on every floor of every office. Scanning opens a Progressive Web App (PWA) — no app installation required.
- b) **Azure AD authentication:** employees log in with their Bayer CWID or corporate email; all personal details (name, email, mobile, department) are auto-populated from Azure Active Directory. Zero manual data entry for employee identity.
- c) **AI-powered ticket structuring:** the employee types their complaint in plain English; the LLM structures it into a standardised, compliant ticket format, auto-categorises the issue type, assigns priority (P1/P2/P3), detects duplicates, and routes the ticket to the correct maintenance team — all before the employee leaves the screen.
- d) **Seven-stage tracked lifecycle:** every ticket moves through **Submitted** → **Acknowledged** → **Assigned** → **In Progress** → **On Hold** (if blocked) → **Resolved** → **Closed**, with timestamps at every stage, visible to the complainant on their device.
- e) **SLA enforcement with auto-escalation:** breach of SLA triggers automatic escalation to the next management tier with full notification trail.
- f) **Multi-channel notification:** email, WhatsApp Business API (two-way actionable), and Microsoft Teams adaptive cards for both employees and facility staff.
- g) **Feedback via Teams:** on resolution, a Teams adaptive card requests a 5-star rating; a web fallback link is surfaced after 10 minutes; the feedback window closes silently after 45 minutes.
- h) **Feedback analytics:** AI-generated reports by day, month, and year — resolution time,

satisfaction scores, category breakdown, worst-performing zones, and technician performance.

1.4 Primary Users & Personas

Persona	Access Channel	Primary Actions
Employee (Complainant)	PWA via QR scan	Raise ticket, track status, provide feedback
Technician / Field Staff	WhatsApp + PWA	Receive assignment, accept via WhatsApp, update progress
Facility Manager	Admin Dashboard	View and manage office tickets, update status, add notes
Operations Head	Admin Dashboard	Cross-office analytics, SLA breach view, escalation management
Super Admin	Admin Dashboard	Office/floor/zone/QR management, user roles, notification config
Internal Audit	Read-only dashboard	Full ticket audit trail, feedback records, SLA compliance view

2 QR Code Architecture

2.1 Location-Aware QR Strategy

Every QR code encodes a unique, structured **Location ID** of the form:

Location ID Format

BAYER-{OFFICE_CODE}-{FLOOR}-{ZONE_CODE}

Examples:

BAYER-THN-03-CAFE -- Thane Office, Floor 3, Near Cafeteria

BAYER-THN-05-CONF-B -- Thane Office, Floor 5, Near Conference Room B

BAYER-MUM-02-RECEP -- Mumbai Office, Floor 2, Reception Area

When an employee scans a QR code, the system automatically pre-fills:

- Office name and city
- Floor number
- Zone / area name

The employee does not type their location. It is locked from the QR metadata. This eliminates a major source of misrouting and removes the need for optional location text boxes entirely.

2.2 QR Generation on Admin Dashboard

The Super Admin generates QR codes from a dedicated page in the admin dashboard:

1. Select office → select floor → select or create zone.
2. System generates a **signed URL**: `https://facilitydesk.bayer.in/raise?loc=BAYER-THN-03-CAFE&sig=`
3. The QR code is rendered on screen with the Bayer logo embedded in the centre (SVG-based, print-ready at A4 and A5 sizes).
4. Admin can download as PDF with a pre-formatted printable card showing the zone name, floor, office, and a short URL below the QR for accessibility.
5. All generated QRs are listed in the admin dashboard with creation date, location, number of tickets raised, and a status toggle (active / retired).

2.3 Signed URL Security Model

- Each QR URL is signed with an HMAC-SHA256 signature using a server-side secret key. The backend verifies the signature on every scan before rendering the complaint form.
- A forged or manually constructed URL without a valid signature is rejected with HTTP 403.
- QR codes can be **retired** by the Super Admin (e.g., when a zone is reconfigured). Retired QR URLs return a friendly message: “This QR code has been decommissioned. Please contact your facility manager.”
- No QR code is valid in perpetuity without being registered in the system — prevents stale printed codes from creating orphaned tickets.

3 Authentication — Azure Active Directory

3.1 CWID / Email Login Flow

1. Employee scans QR → PWA opens in mobile browser.
2. PWA redirects to **Microsoft Identity Platform** (Azure AD OAuth2 / OIDC flow).
3. Employee enters Bayer CWID or corporate email + password on the Microsoft login screen (standard Bayer SSO page — not a custom login form, never touching credentials).
4. On successful authentication, Azure AD returns an ID token and access token to the PWA.
5. PWA backend validates the token, extracts the user’s Object ID, and calls **Microsoft Graph API** to fetch: display name, email, mobile phone, department, office location, and manager.
6. All fetched fields are pre-populated on the complaint form and locked from editing. The employee sees their identity confirmed before submitting.

3.2 Profile Auto-Population from Azure AD

Field	Source	Used For
Display Name	Azure AD Graph	Shown on ticket, notification addressee
Corporate Email	Azure AD Graph	Email notifications, Teams message target
Mobile Number	Azure AD Graph	WhatsApp notification to employee
Department	Azure AD Graph	Ticket metadata; routing context
Office Location	Azure AD Graph	Cross-validation with QR location
Manager	Azure AD Graph	Escalation contact for P1 tickets

3.3 Role-Based Access Control

Roles are managed as **Azure AD App Roles** assigned to security groups:

- **Employee** — raise tickets, view own tickets, provide feedback.
- **Technician** — view assigned tickets, update status, add resolution notes.
- **FacilityManager** — full view of tickets for their office, all status transitions, add internal notes.
- **OperationsHead** — cross-office view, analytics, SLA breach management.
- **SuperAdmin** — QR management, office/floor/zone config, user role assignment, notification config.
- **Auditor** — read-only access to all tickets and audit logs.

4 Ticket Submission & AI Structuring

4.1 Employee-Facing Web App (PWA)

The complaint form presented after authentication contains:

- **Pre-filled and locked:** Employee name, email, department, office, floor, zone (from Azure AD + QR metadata).
- **Plain-English complaint box:** a single large text area with the placeholder “Describe your

- issue in your own words...*” Character limit: 1000. No dropdowns, no mandatory category selection — the employee just types.
- **Optional photo upload:** up to 3 images (JPEG/PNG, max 5 MB each). Stored in Azure Blob Storage; linked to the ticket.
 - **Submit button:** triggers the AI structuring pipeline and shows a live “Processing your request...” animation before displaying the confirmed ticket.

4.2 AI Categorisation, Prioritisation & Auto-Routing

On submission, the raw complaint text is sent to the AI pipeline:

1. **Categorisation:** LLM classifies the complaint into one of the predefined taxonomy categories:

Category	Category	Category	Category
Plumbing	Electrical	HVAC / AC	Furniture
Housekeeping	IT Infrastructure	Safety	Lift / Elevator
Pantry / Kitchen	Pest Control	Access / Security	Other

2. **Priority assignment:** LLM assigns one of three priority levels:

Priority	Definition	Example
P1	Safety risk or total blocker	“Exposed live wire near workstation”, “Lift stuck with person inside”
P2	Affects productivity	“AC not working in entire wing”, “Washroom out of order”
P3	Comfort / non-urgent	“Chair is wobbly”, “Light flickering in corridor”

3. **Auto-routing:** ticket is assigned to the maintenance team responsible for the detected category. The routing table is maintained by the Super Admin in the admin dashboard and maps each category to one or more responsible teams per office.
4. **Structured ticket generation:** LLM rewrites the complaint into a standardised format (see §4.4). The same JSON schema is enforced across all tickets regardless of how the complaint was phrased.

4.3 Duplicate Detection & Ticket Merging

- On submission, the system checks for open tickets from the same zone with the same category raised in the last 24 hours.

- If a semantic match is found (using vector similarity on the complaint text embeddings), the new submission is **merged** into the parent ticket as a “+1 affected employee” entry.
- The employee sees: “*Your issue has been grouped with a similar report from your zone. Ticket #XXXX is already being attended to.*” with a link to track that ticket.
- The facility manager sees the merged ticket with a count of affected employees — useful for prioritising truly widespread issues.

4.4 Structured Ticket Format

Every ticket stored in the database and shown to facility managers follows an identical, LLM-enforced schema:

Standard Ticket Schema (JSON)

```
{
  "ticket_id": "BYR-THN-2026-003847",
  "status": "Submitted",
  "priority": "P2",
  "category": "HVAC / AC",
  "location": { "office": "Thane", "floor": 3, "zone": "Cafeteria" },
  "summary": "Air conditioning unit non-functional",
  "description": "The AC unit in the cafeteria area on Floor 3
    has not been operational since morning. Multiple
    employees are affected. Immediate attention required.",
  "reported_by": { "name": "...", "email": "...", "dept": "..." },
  "created_at": "2026-05-25T09:14:33Z",
  "assigned_team": "HVAC-Maintenance-Thane",
  "sla_deadline": "2026-05-25T17:14:33Z"
}
```

5 Ticket Lifecycle & Escalation Engine

5.1 Seven-Stage Workflow

Ticket Lifecycle Stages

Submitted → Acknowledged → Assigned → In Progress → On Hold →
Resolved → Closed

Stage	Trigger	What Happens
Submitted	Employee submits	Ticket created; AI structures it; notifications sent to team
Acknowledged	Admin or auto (2 min)	System confirms receipt; employee notified; SLA clock starts
Assigned	Facility Manager assigns	Technician name attached; technician notified via WhatsApp + email
In Progress	Technician accepts	Technician on-site; employee notified with ETA if available
On Hold	Technician raises block	Reason required (e.g., “waiting for spare part”); SLA paused; manager alerted
Resolved	Technician marks resolved	Resolution note mandatory; employee notified; feedback flow triggered
Closed	48 hours after Resolved	Auto-closes if no dispute raised; feedback window closed

The employee sees a **visual stepper** on their device — identical in concept to a package delivery tracker — showing all seven stages with timestamps filled in as the ticket progresses.

5.2 SLA Definitions per Priority

Priority	Acknowledge By	Resolve By	Escalation Path
P1	15 minutes	2 hours	Immediate: Facility Manager + Operations Head + Manager of complainant
P2	30 minutes	8 hours	Facility Manager at 4 hours; Operations Head at 6 hours
P3	2 hours	48 hours	Facility Manager at 24 hours

5.3 Automatic Escalation Rules

The escalation engine runs as a scheduled Celery task every 5 minutes:

1. Queries all open tickets where the current stage’s SLA deadline has passed.
2. For each breached ticket: moves stage to **Escalated** sub-status (displayed in **red** on the

dashboard); sends escalation notification to the next tier.

3. **P1 escalation path:** after 15 min with no acknowledgement → notify Facility Manager via WhatsApp + email; after 30 min → notify Operations Head; after 1 hour → notify Head of Facilities (Gandharv Chauhan) directly with full ticket detail.
4. **P2/P3 escalation:** follows the SLA table above; each escalation adds a timestamped entry to the ticket audit trail.
5. The escalation audit trail is immutable — every escalation event, who was notified, and at what time is stored permanently with the ticket.

5.4 Admin Ticket Management View

The facility manager's ticket view shows:

- Kanban board with columns per lifecycle stage; cards draggable to advance status.
- SLA countdown timer on each card; turns **amber** at 75% elapsed, **red** at breach.
- Filter by office, floor, zone, category, priority, technician, and date range.
- Click-through to full ticket detail: employee info, AI-generated structured description, original raw complaint, photo attachments, full stage history with timestamps, escalation log, and resolution notes.
- **Bulk assign:** select multiple unassigned tickets and assign to a technician in one action.
- **Merge:** manually merge two tickets the system did not auto-detect as duplicates.

6 Notification System

6.1 Email Notifications (Azure Communication Services)

- **To employee:** ticket confirmation (with ticket ID and tracker link), status change notifications, resolution confirmation, feedback request link.
- **To facility team:** new ticket alert with full structured detail and direct link to admin dashboard ticket.
- **To escalation contacts:** breach notification with ticket summary, time elapsed, and direct resolve link.
- All emails are templated with Bayer branding; templates managed in the admin dashboard.

6.2 WhatsApp Business API — Two-Way Actionable Messages

WhatsApp messages are sent via the **WhatsApp Business API** (Meta Cloud API or Twilio) using pre-approved message templates:

New ticket assigned to technician (template: TICKET_ASSIGNED):

“FacilityDesk | New Ticket #BYR-THN-2026-003847

Category: HVAC / AC | Priority: P2 | Location: Thane, Floor 3, Cafeteria

Summary: Air conditioning unit non-functional

Please respond: [Accept] [Escalate]"

Tapping Accept: sends a webhook to FastAPI → ticket status moves to **Acknowledged**; employee notified automatically.

Tapping Escalate: sends a webhook → ticket escalated to Facility Manager; reason requested via follow-up WhatsApp message.

- Technicians can also update status via WhatsApp reply commands: **IN-PROGRESS**, **ON-HOLD**: **reason**, **RESOLVED**: **resolution note**.
- All WhatsApp interactions are logged to the ticket audit trail with timestamp and message content.
- Employee also receives a WhatsApp confirmation when ticket is raised and when resolved.

6.3 Microsoft Teams Adaptive Cards

- Facility Manager teams channel receives a formatted adaptive card for every new P1 and P2 ticket raised in their office.
- Operations Head channel receives daily SLA breach summary at 6 PM.
- **Feedback card** sent to employee on resolution: 5-star rating buttons + one optional free-text line, directly actionable inside Teams without opening a browser.

7 Feedback Collection System

7.1 Two-Touch Feedback Flow

1. **Trigger:** ticket moves to **Resolved**.
2. **Touch 1 (T+0 min):** Microsoft Teams adaptive card sent to employee's Teams account with: 5-star rating buttons, one optional free-text field ("Tell us more..."), and the ticket summary. Actionable inside Teams.
3. **Touch 2 (T+10 min):** if no feedback received, a feedback link appears as a banner notification inside the FacilityDesk PWA when the employee next opens it. The link opens a minimal one-page form.
4. **Window close (T+45 min):** if still no feedback, the feedback request is silently closed. No further nudges are sent. The ticket moves to **Closed** status.
5. Feedback is never mandatory. Employees are never penalised for not providing it.

7.2 Feedback Storage & Linkage

Every feedback response is stored with:

- Ticket ID, ticket category, office, floor, zone.

- Assigned technician ID and name.
- Rating (1–5 stars), free-text comment (if provided).
- Submission channel (Teams / PWA web form).
- Time from ticket creation to resolution (used for SLA analytics).
- Submission timestamp.

8 End-to-End Data Pipeline & Architecture

8.1 Architecture Diagram

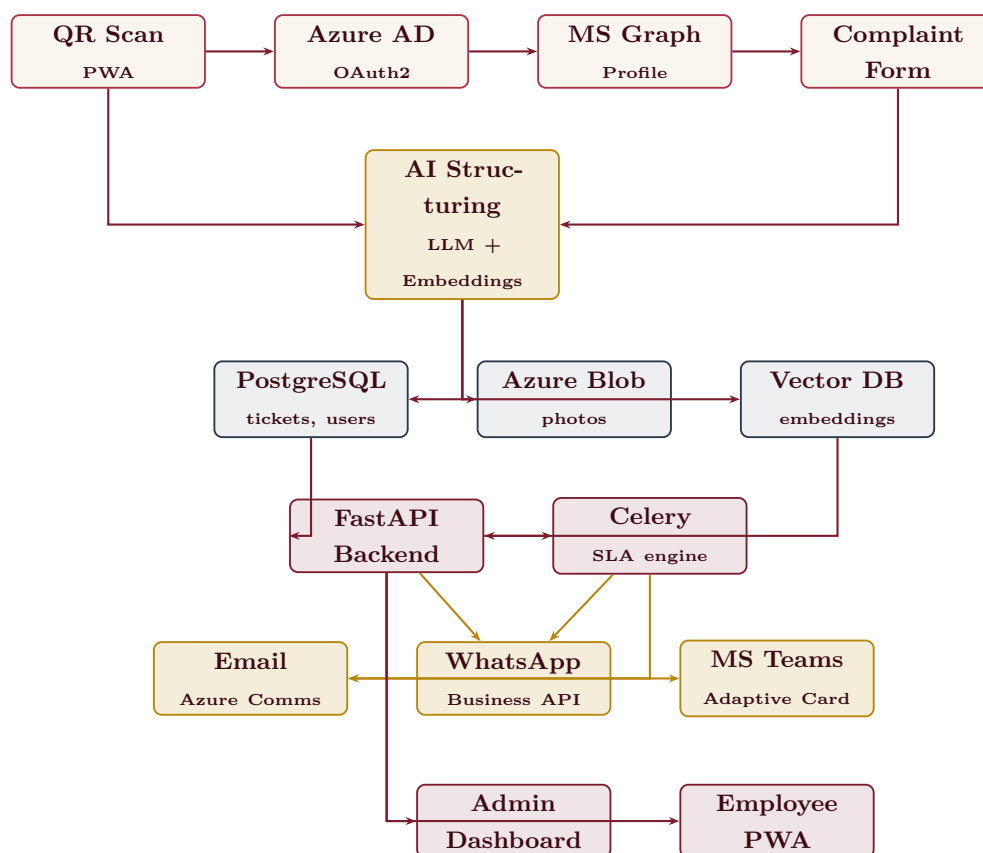


Figure 1: FacilityDesk end-to-end platform architecture.

8.2 Pipeline Steps 0 through 6

Step 0 — QR Scan & Auth: Employee scans QR. PWA opens. Signed URL validated (HMAC-SHA256). Azure AD OAuth2 login. Microsoft Graph API call fetches profile. Complaint form pre-populated.

Step 1 — Submission & AI Structuring: Raw complaint text + optional photos submitted. LLM pipeline: categorise → assign priority → check duplicates (vector similarity) → generate structured ticket JSON → determine assigned team from routing table.

Step 2 — Persistence: Structured ticket written to PostgreSQL. Photos stored in Azure Blob Storage with ticket ID reference. Complaint embedding stored in Vector DB for future duplicate detection.

Step 3 — Notification Dispatch: FastAPI triggers notification service: email to employee (confirmation) + email to assigned team + WhatsApp to technician (with Accept/Escalate buttons) + Teams card to facility manager channel (P1/P2 only).

Step 4 — Lifecycle Management: Facility manager updates ticket status via admin dashboard Kanban. Technician updates via WhatsApp commands or PWA. Each status change: timestamped in PostgreSQL, notification sent to employee, audit trail entry written.

Step 5 — SLA & Escalation Engine: Celery Beat runs every 5 minutes. Queries tickets approaching or past SLA deadline. Triggers escalation notifications to next tier. Updates ticket escalation log. Sends breach summary to Operations Head channel at 6 PM daily.

Step 6 — Resolution & Feedback: Technician marks **Resolved** with mandatory resolution note. Employee notified. Teams feedback card sent (T+0). Web fallback banner at T+10 min. Feedback window silently closed at T+45 min. Ticket auto-moves to **Closed** after 48 hours.

8.3 Dependency Graph

```
QR Scan → Azure AD Auth → MS Graph Profile fetch
→ Complaint Form (pre-filled)
→ LLM AI Pipeline (categorise, prioritise, deduplicate, structure)
→ PostgreSQL (ticket record)
→ Vector DB (duplicate detection embedding)
→ Azure Blob (photo attachments)
→ FastAPI Notification Service
→ Email (employee + team)
→ WhatsApp Business API (technician — two-way)
→ Microsoft Teams (manager channel + feedback card)

Celery SLA Engine (every 5 min)
→ queries PostgreSQL for breached tickets
→ escalation notifications (Email + WhatsApp)
→ updates ticket escalation audit trail
```

9 AI / LLM Role & Boundaries

9.1 Deterministic Logic — No LLM Needed

- Signed URL HMAC validation.
- Azure AD token validation and Graph API calls.
- SLA deadline calculation and breach detection (Celery).

- Status transition enforcement (only valid transitions allowed in the state machine).
- Notification dispatch routing (which channel, which recipient).
- Feedback window timer (T+10, T+45).
- All PostgreSQL read/write operations.
- QR code generation (deterministic from location ID).

9.2 LLM-Assisted Tasks & Prompt Sketches

Task	LLM Input	LLM Output
Ticket structuring	Raw complaint text + location + employee dept	Structured JSON: summary, description, category, priority
Duplicate detection	New complaint embedding vs. open ticket embeddings	Match score + parent ticket ID if duplicate
Feedback report generation	Aggregated feedback data for period	Narrative AI report with insights, trends, recommendations
Anomaly commentary	Spike in ticket volume for a zone	Natural-language explanation surfaced on admin dashboard

Ticket structuring prompt:

```
// SYSTEM
You are a facilities management ticket assistant for Bayer offices.
Given an employee’s complaint, return ONLY valid JSON matching
the standard ticket schema. Never deviate from the schema.
Categories: Plumbing, Electrical, HVAC/AC, Furniture,
Housekeeping, IT Infrastructure, Safety, Lift/Elevator,
Pantry/Kitchen, Pest Control, Access/Security, Other.
Priorities: P1 (safety/total blocker), P2 (productivity),
P3 (comfort/non-urgent).

// USER
Complaint : "{raw_complaint}"
Department: {employee_dept}
Location : {office}, Floor {floor}, Zone: {zone}
```

Consistency enforcement: The LLM is instructed to return the same category and priority for semantically identical complaints regardless of how they are phrased. The system prompt includes 3 few-shot examples per category to anchor classification. If LLM confidence is below threshold (parsed from logprobs or a self-rated confidence field), the ticket is flagged for facility manager review before routing.

10 Admin Dashboard Design

10.1 Three Admin Personas

- **Facility Manager** (per office): sees only their office. Full ticket management, status updates, technician assignment, resolution notes. Cannot see other offices.
- **Operations Head**: cross-office view. Analytics, SLA breach tracker, escalation management, technician performance. Cannot directly edit tickets in other offices.
- **Super Admin**: all of the above + QR management, office/floor/zone config, routing table, notification templates, user role assignment, system settings.

10.2 Page Structure

- `/admin/dashboard` — live ticket Kanban; SLA countdown; open/in-progress/overdue counts per office.
- `/admin/tickets` — full table view with advanced filters (date, office, floor, category, priority, technician, status).
- `/admin/tickets/{id}` — ticket detail: full lifecycle timeline, AI-structured vs raw complaint side-by-side, photo attachments, escalation log, resolution note, feedback received.
- `/admin/analytics` — cross-office charts: tickets by category, by office, by priority; average resolution time; SLA compliance rate; repeat-issue zones.
- `/admin/feedback` — feedback report page (see §10.4).
- `/admin/qr` — QR generation and management (see §10.3).
- `/admin/config` — routing table, notification templates, SLA thresholds, office/floor/zone management, user roles.

10.3 QR Generation & Print Module

- **Generate QR**: Super Admin selects office → floor → zone; clicks Generate. System creates location ID, signs the URL, renders QR (SVG with Bayer logo watermark centred), previews on screen.
- **Print-ready card**: A4 and A5 PDF download; includes Bayer branding, zone name in large print above the QR, short URL below, “Scan to report a facility issue” instruction in English and Hindi.
- **QR inventory table**: all generated QRs listed with location, creation date, total tickets raised from this QR, last ticket date, active/retired status toggle.
- **Bulk generation**: Super Admin can generate QRs for all zones of a floor in one batch operation; download as a ZIP of print-ready PDFs.

10.4 Feedback Report Generator

The feedback report page allows the Operations Head or Super Admin to generate an AI-produced report:

1. Select time period: daily / monthly / yearly (or custom date range).
2. Select scope: all offices / specific office / specific floor.
3. Click **Generate Report**. LLM receives aggregated feedback data and produces a narrative report including:
 - Overall satisfaction score and trend vs. previous period.
 - Category breakdown: which issue types have the lowest satisfaction scores.
 - Worst-performing zones: offices/floors with highest ticket volume and lowest ratings.
 - Best and lowest-rated technicians (anonymised in external-facing views).
 - Average time to resolve per priority and per category.
 - Notable recurring issues: zones with the same category ticket appearing 3+ times in the period.
 - Recommendations: LLM-generated actionable suggestions (e.g., “Floor 3 AC has been reported 7 times this month — preventive maintenance contract recommended”).

Report is downloadable as a formatted PDF and viewable on-screen.

11 System Architecture & Technology Stack

Layer	Technology	Rationale
Employee Frontend	Next.js PWA	Progressive Web App; no install required; works on any mobile browser after QR scan
Admin Dashboard	Next.js + MUI	Server-side rendering for large ticket tables; role-aware routing
Auth	Azure AD + MSAL.js	Corporate SSO; Bayer CWID login; Microsoft Graph API for profile data
Backend API	FastAPI (Python 3.12)	Async REST; WebSocket for live Kanban updates; Celery task dispatch
AI / LLM	Azure OpenAI GPT-4o	Ticket structuring, categorisation, duplicate detection, feedback reports
Embeddings	Azure OpenAI Ada-002	Complaint text vectorisation for semantic duplicate detection
Vector DB	pgvector (PostgreSQL)	Semantic similarity search for duplicate detection; no separate infra needed
Operational DB	PostgreSQL 15	Tickets, users, feedback, audit trail, routing config, QR registry
File Store	Azure Blob Storage	Photo attachments, generated PDFs, QR SVGs
Scheduling	Celery + Redis Beat	SLA engine (every 5 min); feedback window timer; daily SLA summary
Email	Azure Communication Services	Transactional email with Bayer-branded templates
WhatsApp	Meta WhatsApp Business API	Two-way technician notifications; employee status updates
Teams	Microsoft Teams Webhook + Bot	Adaptive cards for managers; feedback collection
QR Generation	Python <code>qrcode</code> + <code>cairosvg</code>	SVG QR with Bayer logo; PDF print-ready export
Hosting	Azure App Service	Aligned with Bayer's existing Azure tenant

12 Security & Compliance

- **Authentication:** 100% Azure AD; no custom credential storage; MSAL handles token lifecycle.
- **QR security:** HMAC-SHA256 signed URLs; secret key rotated quarterly; retired QRs immediately invalidated server-side.
- **Data encryption:** all data at rest encrypted via Azure Storage Service Encryption (AES-256); all traffic over TLS 1.2+.
- **PII handling:** employee name, email, and mobile stored in PostgreSQL with column-level access restricted to **FacilityManager** role and above for the name/department fields; mobile number masked in public audit views.
- **Role enforcement:** all API endpoints validate the Azure AD App Role claim from the JWT before returning any data. A **Technician** token cannot access the admin analytics endpoint.
- **WhatsApp message integrity:** webhook callbacks from Meta are validated using the **X-Hub-Signature-256** header before processing any status updates.
- **Audit trail:** every ticket status change, escalation event, notification sent, and feedback received is written to a PostgreSQL `audit_log` table with user ID, timestamp, and action. Rows are insert-only (no updates or deletes permitted).
- **Data retention:** ticket and feedback data retained for 3 years per Bayer's internal records policy; personal data pseudonymised after 1 year.
- **GDPR / PDPB compliance:** employees can request deletion of their personal data via a self-service form; pseudonymisation is applied within 30 days of request, preserving aggregate analytics without identifiable data.

13 Evaluation Metrics

Metric	Target
Ticket categorisation accuracy	Precision $\geq 95\%$ vs. manually labelled benchmark of 500 tickets
Priority assignment accuracy	Precision $\geq 97\%$ for P1 tickets (safety-critical; zero tolerance for mis-classification)
Duplicate detection	Precision $\geq 90\%$, Recall $\geq 85\%$ on same-zone same-category open tickets
P1 acknowledge time	100% of P1 tickets acknowledged within 15 minutes
P2 resolution SLA	$\geq 90\%$ of P2 tickets resolved within 8 hours
Feedback response rate	Target $\geq 40\%$ within the 45-minute feedback window
Ticket submission time	Employee completes scan-to-submit in under 90 seconds (including auth)
Admin dashboard load	≤ 2 seconds for Kanban board with up to 500 open tickets
Notification delivery	WhatsApp and email delivered within 30 seconds of trigger event
System uptime	$\geq 99.5\%$ during business hours (8 AM – 8 PM IST, all offices)

14 Implementation Timeline

Estimated for a team of 1 backend engineer, 1 frontend engineer, and 1 LLM/AI engineer. Timeline is structured to deliver the QR-to-ticket flow and admin Kanban first (highest immediate value), followed by notifications, escalation engine, and analytics.

Period	Deliverables
Weeks 1–2	Azure AD app registration; Microsoft Graph API access; PostgreSQL schema; QR generation module; location ID taxonomy for all offices and floors.
Weeks 3–4	Employee PWA: QR scan flow, Azure AD OAuth2 login, profile auto-population, complaint form, photo upload, ticket submission to backend.
Weeks 5–6	AI pipeline: LLM ticket structuring, categorisation, priority assignment, pgvector embedding store, duplicate detection, routing table config.
Weeks 7–8	Admin dashboard: Kanban board, ticket detail view, status update flow, technician assignment, filters and search, audit trail view.
Weeks 9–10	Notification system: Azure Communication Services email, WhatsApp Business API integration (two-way webhook), Microsoft Teams adaptive cards and feedback card.
Weeks 11–12	SLA engine: Celery Beat task, breach detection, automatic escalation notifications, escalation audit trail, SLA countdown on Kanban cards.
Weeks 13–14	Feedback collection: Teams card trigger on resolve, T+10 PWA banner fallback, T+45 silent close, feedback storage and linkage to ticket.
Weeks 15–16	Analytics and feedback report: cross-office charts, AI-generated period report (daily/monthly/yearly), PDF export, QR inventory with ticket-count analytics.
Weeks 17–18	Security hardening: role enforcement audit, HMAC QR validation, PII masking, audit log insert-only enforcement, GDPR self-service deletion flow.
Weeks 19–20	Pilot deployment at Thane office (2 floors); UAT with facility team; WhatsApp template approval (Meta review); bug fixes and documentation handover.
Total	≈ 20 weeks (≈ 5 months) to pilot-ready platform, with full rollout across all offices in the subsequent 4–6 weeks.

15 Future-Proofing Architecture

15.1 Configuration-First Principle

Core Design Rule

No office, floor, zone, category, SLA threshold, routing rule, or notification template should be hardcoded. All of these live in PostgreSQL config tables editable by the Super Admin through the admin UI. Adding a new office, a new floor, a new issue category, or changing an SLA target requires zero code changes and zero redeployment.

15.2 New Office Onboarding

Adding a new Bayer office location requires only:

1. Super Admin adds office record (name, city, address) via admin UI.
2. Adds floors and zones for that office.
3. Generates QR codes for each zone (batch operation).
4. Assigns a Facility Manager Azure AD account to the office.
5. System automatically includes the new office in all analytics, escalation paths, and reports.

Zero code change. Zero redeployment. New office is live in under 30 minutes.

15.3 Extensibility Roadmap

Future Addition	Effort	Implementation Path
New office	Negligible	Admin UI config; QR batch generation
New issue category	Low	Add row to category config table; update LLM system prompt
Native mobile app	Medium	React Native reuse of PWA components and same FastAPI backend
SAP PM integration	Medium	FastAPI connector to SAP Plant Maintenance OData for work-order creation on ticket resolution
Predictive maintenance	High	Train ML model on historical ticket data to predict equipment failure before it occurs
Multi-language support	Low	Add language detection to LLM prompt; i18n in Next.js frontend

Document Control. This document is version 1.0. It should be reviewed after the Thane pilot (Weeks 19–20) and updated to reflect lessons from the first deployment. WhatsApp Business API template approvals from Meta must be initiated no later than Week 8 to avoid delaying the notification system delivery in Week 9–10. Azure AD app registration and Microsoft Graph API consent must be confirmed with Bayer IT before Week 1 development begins.